2021 Ph H2 Q7

Section: Particles and Waves

Topic: Nuclear Reactions

This question deals with energy released during a nuclear reaction.

(a)
Calculate the energy released in MeV

The nuclear reaction involves a change in mass (mass defect). The energy released is given by:

$$E = \Delta m c^2$$

Given:

- $\Delta m = 0.020\,\text{u}$
- $1\,\text{u} = 931.5\,\text{MeV}$

$$E = 0.020 \times 931.5 = 18.6\,\text{MeV}$$

(b)
Calculate the total energy released by a sample containing 1.0×10^{20} reactions

$$E_{\text{total}} = 18.6 \times 10^{20}\,\text{MeV}$$

Convert MeV to joules:
$1\,\text{MeV} = 1.6 \times 10^{-13}\,\text{J}$

$$E = 18.6 \times 10^{20} \times 1.6 \times 10^{-13} = 3.0 \times 10^{9}\,\text{J}$$

🧠 Revision Tips

- Use the conversion:
$1\,\text{u} = 931.5\,\text{MeV} \quad \text{and} \quad 1\,\text{MeV} = 1.6 \times 10^{-13}\,\text{J}$
- Energy from nuclear reactions is **mass converted into energy**
- Multiply energy per reaction by the **number of reactions** for total energy
- Always keep track of units: MeV vs J
